

E-009 OPEN ITEMS / FIELD VERIFICATION

Conv_Simple / CV-101 (garage conveyor) — Generated from open_items.yaml — the formal field-verification docket

OPEN ITEMS (generated)

28 open items; 2 resolved

OI	Sheet	Item	Verify
OI-01	E-005	+24 VDC source terminal on PS1 feeding the input rail	Meter PS1 +24V to 0V = 24 VDC; confirm which terminal feeds the input devices.
OI-02	E-005	Input common (COM0) grouping + sink vs source jumpering on the Micro820	Confirm which embedded input common serves I-00..I-05 and whether inputs are sink or source; meter COM0 to 0V.
OI-03	E-005	All input wire numbers (W200..W205, W24, W0V are PROPOSED)	Read the actual wire markers on the panel; replace proposed numbers with as-built.
OI-04	E-005	Field-device terminal designations (S0 11-12/23-24, SS1 FWD/REV/COM, S2 3-4, B1 BN/BU/BK)	Confirm each device's real terminal markings against its nameplate/datasheet.
OI-05	E-005	E-stop architecture between S0 and the PLC	Confirm whether S0 lands directly on I-02/I-03 or via a safety relay; a monitored input is NOT a safety stop (see safety note).
OI-06	E-005	Photo-eye B1 device (historically 'unmapped') — make/model, PNP/NPN, powered source	Identify the physical sensor; confirm BN/BU/BK to +24V/0V/I-05; PNP sourcing assumed.
OI-07	E-005	Direction selector SS1 physical device (2-pos vs 3-pos FWD-OFF-REV)	Confirm selector type and that FWD/REV contacts map to I-00/I-01.
OI-08	E-005	Spares I-06..I-11 confirmed unused	Confirm no field wires land on I-06..I-11 (program shows none).
OI-09	E-006	Output bank technology + common feeds. +CM0/-CM0/+CM1/-CM1 polarity naming suggests DC transistor banks; devices.yaml io line says 'relay DO' — conflict.	Resolve per 2080-IN009 + meter the bank feed (expect PS1 +24V/0V).
OI-10	E-006	Q1 contactor identity — part number, coil voltage, pole count; coil terminal ids A1/A2 are PROPOSED.	Meter coil circuit from O-02.
OI-11	E-006	PL1/PL2 devices + lamp terminal ids (X1/X2 proposed); S2 lamp terminals.	Meter the light circuits and confirm device markings.
OI-12	E-006	Confirm no field wires on spare O-04..O-06 (+CM1 bank unused).	Trace the PLC output block and verify no wires to O-04, O-05, O-06.
OI-13	E-005, E-006	Re-confirm input AND output map against the live Conv_Simple_1.8 CCW project; v4.0 vintage drift proven on I-05 'Entry sensor' vs live photo-eye.	Open the live Conv_Simple_1.8 CCW project and cross-check every I-xx and O-xx assignment against the drawn functions (I-05 drift already proven; O-02 has independent Prog_init corroboration).
OI-14	E-003	Q1 contactor placement in the power chain (assumed line side of VFD1 per 'safety power'; could be elsewhere).	Trace conductors on the bench; verify Q1 poles 1-6 actual locations. R-C surge absorber both ends recommended per GS10 manual.

OI	Sheet	Item	Verify
OI-15	E-003	Supply voltage/receptacle + CB1 existence/type/rating (2-pole, single-phase 230 V branch breaker).	Identify the bench supply; confirm CB1 device and rating. NEC and GS10 manual require upstream protection.
OI-16	E-003	GS10 exact model+frame (enables wire-gauge/fuse tables); M1 nameplate; conductor gauges; PE points + ground resistance ≤0.1Ω check.	Photograph GS10 nameplate and motor nameplate; measure PE resistance between drive/motor grounds and supply PE.
OI-17	E-003	RFI jumper in/out per grounding topology (symmetrical vs corner-grounded unknown).	Determine bench PE bonding strategy; verify RFI jumpers per GS10_UM.txt L1693-1718 if present.
OI-18	E-006	Fallback DI wiring physical presence (expected ABSENT — P02.0x factory default).	Trace the PLC output block; confirm zero hardwired leads to any GS10 digital inputs (fallback path should be unpopulated).
OI-19	E-003, E-005, E-006	Wire-numbering convention adopted: W[sheet-digit][line-2d] (see wires.yaml convention + the E-001 key). E-005 renumbered W200..W205 → W500..W505. Rails (W24, W0V) and E-007 mnemonic labels (485+/485-/SGND/SH) are exempt. All numbers PROPOSED until as-built.	Read the actual wire markers on the panel; replace proposed W-numbers with as-built (mirror of OI-03).
OI-20	E-007	GS10 comms line params: 2026-05-20 export = 38.4k/8N2 vs 2026-05-26 bench sniff = 9600/8N1 — adjudicate.	Fresh keypad readback of the GS10 serial config; update print if params differ from current 9600/8N1 assumption.
OI-21	E-003	RESOLVED (2026-07-11): no separate motor contactor — the MLC control relay IS the single-phase drive-supply switch (2 NO contacts). See E-003.	Confirm whether a separate motor contactor exists between CB1 and VFD1; the device labeled 'MLC' (Q1) is a Schneider TeSys CA3KN22BD CONTROL RELAY — it has no main power poles.
OI-22	E-006, E-007	GS10 control map (full-res photo): top run/DI connector (FWD/REV/DI3-5/+24V/DCM) confirmed EMPTY; bottom analog/output row (+10V/ACM/AI/AO1/DO1/DOC) confirmed WIRED. Identify which bottom-row terminals are used (speed reference +10V/AI/ACM vs feedback AO1/DO1/DOC) and to what. Confirm nothing on the top run/DI row.	Map which of FWD/REV/DI3/DI4/DI5/AI/AO1/DO1/DOC are landed and to what; reconcile hybrid Modbus + hardwired-terminal control.
OI-23	panel	Siemens CPU 1212C AC/DC/RLY (S7-1200) observed on the panel (photo wire_2) — role vs CV-101 unknown.	Confirm whether the Siemens CPU 1212C controls any CV-101 function before referencing it on any CV-101 sheet; not drawn until evidenced.
OI-24	panel	Device labeled "PMC 192.x" observed on the panel (photo wire_2).	Identify the device (network device / meter?) and its relationship, if any, to CV-101.
OI-25	E-004	DC +/- distribution block: which color (blue vs white) is +24V vs 0V, circuit count, PS1 branch fusing; PLUS PS1's AC input source — which leg/circuit feeds L and N (120 V leg vs the 230 V 1ϕ supply) is not documented.	Trace/meter the distribution block conductors against PS1 +V/-V output; confirm color-to-polarity mapping, circuit count, and any branch fusing (photo wire_1/wire_2). Separately, identify the branch circuit feeding PS1.L/PS1.N and its voltage (single leg of the 230 V 1ϕ supply, or a dedicated 120 V circuit).
OI-26	E-006	MLC (Q1) aux contacts (full-res photo): coil A1/A2 wired + NO contacts 13-14 & 43-44 in use confirmed; NC 21-22/31-32 appear unused. Trace where 13-14 and 43-44 land in the control circuit.	Trace each aux contact to its destination in the control circuit; confirm function (e.g., interlock feedback, seal-in, indicator).
OI-27	E-003	RESOLVED (2026-07-11): motor is 230 V (technician-confirmed) — the GS10 is a standard 230 V 1ϕ-in / 230 V 3ϕ-out drive; the earlier reported 480 V was a misstatement. E-003 letters 230 V 3ϕ output.	Closed by technician confirmation — review/PHOTO_EVIDENCE_V7.md §3 ('it's 230'), not PHOTO_EVIDENCE_V6.md (which predates this confirmation and still shows 480 V/pending). Exact GS10 model/frame still tracked under OI-16 (for wire-gauge/fuse tables).
OI-28	E-004	PS1 enclosure/chassis ground and DC-0V-to-PE bonding not documented — is PS1's DIN-rail/enclosure bonded to PE, and is the 24 VDC 0V rail (DB1 0V-bus) intentionally tied to PE anywhere (SELV/PELV grounding scheme), or left floating?	Meter PS1 enclosure/DIN rail to the PE bus for continuity; meter DB1 0V-bus to the PE bus for continuity/resistance; confirm whether a deliberate 0V-PE bond exists or the 24 VDC system is intentionally floating.

NOTES

- RESOLVED items stay listed (grayed, not deleted) for traceability — see the item text for the resolving evidence and date.
- Closing an open item requires evidence — a repo artifact citation or a bench measurement — not a re-assertion (see the model discipline note on E-001).

SOURCES:

model/open_items.yaml (generated — do not hand-edit this table; re-run emit after any model change)

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Drawn: MIRA / FactoryLM

SHEET
E-009

REV A

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2026-07-11